

DSA210 Project Proposal

This project aims to identify which openings provide the best statistical winning chances for different Elo rating brackets. By quantifying the "win probability" of specific strategies at various levels (e.g., <1200, 1200-1800, 1800+), this research will help players select opening repertoires that align with their actual skill level rather than professional theory.

The primary dataset for this project will be the Chess Game Dataset (Lichess) available on Kaggle, which contains approximately 20,000 games. This dataset includes key features such as `white_rating`, `black_rating`, `opening_name`, `moves`, `winner`, and `victory_status`.

Moreover, this primary dataset will be augmented with data from the Lichess API. This enrichment will involve retrieving an additional 80,000 to 100,000 games using Python-based tools. The goal of this enrichment is to increase the sample size for specific opening variations and to ensure a balanced distribution of games across all rating tiers.

The main goal is to determine the extent to which a player's rating affects the win probability of specific chess openings. The project will involve:

Data Cleaning and Preparation: Parsing move strings and categorizing games into rating-based skill tiers.

Exploratory Data Analysis (EDA): Visualizing win/loss ratios for the most common openings in each skill tier to identify performance shifts.

Statistical Testing: Using hypothesis testing to determine if the success of specific opening types (e.g., aggressive vs. positional) is statistically significant across different rating groups.

Machine Learning: Developing a classification model (such as Random Forest or Logistic Regression) to predict game outcomes based on the opening choice, the rating gap between opponents, and game duration.